

Introduction to Earth System

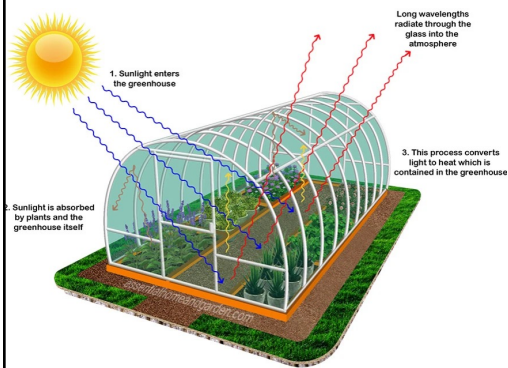
The greenhouse effect

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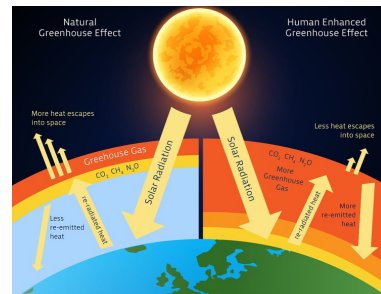
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The real greenhouse



Earth



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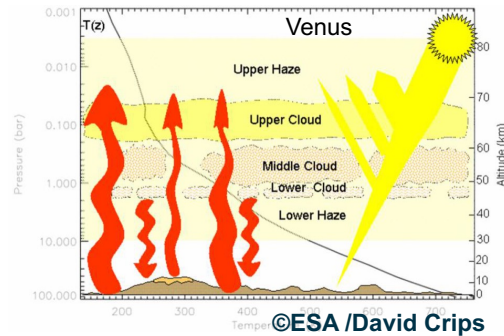
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The greenhouse effect

96.5% CO₂

Runaway greenhouse effect

Venus' oceans may have boiled away in a "runaway greenhouse effect"



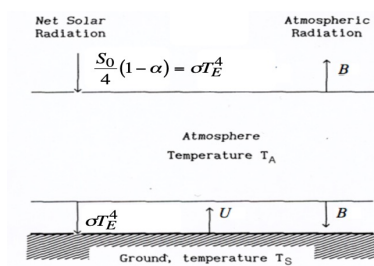
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The Greenhouse Effect

- In 0-D model (the Earth is considered as one point) → $T_E = 255K$
- Case 1: the atm with 1 layer.** Supposing the atm is transparent to **shortwave radiation**; the atm and the ground behave as blackbodies with respect to **longwave radiation**



Write the energy balance equations at:

- TOA
- Atmosphere
- Surface

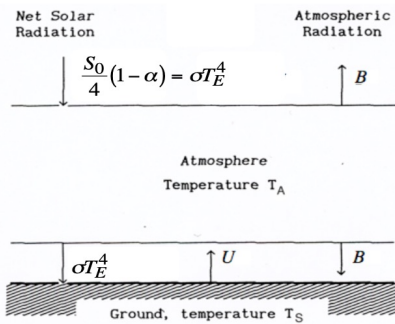
? T_S, T_A

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The Greenhouse Effect



At the TOA

$$\frac{S_0}{4}(1-\alpha) = B \equiv \sigma T_E^4 = \sigma T_A^4$$

For the atmosphere

$$U = \sigma T_S^4 = 2B = 2\sigma T_A^4$$

$$T_S = \sqrt[4]{2} T_E$$

$$T_E = 255\text{K} \rightarrow T_S = 303\text{K}$$

$\rightarrow T_S > 288\text{K}$ (observed value)

$\rightarrow$ The presence of the atm raises the temperature of the Earth $\rightarrow$ **the greenhouse effect**

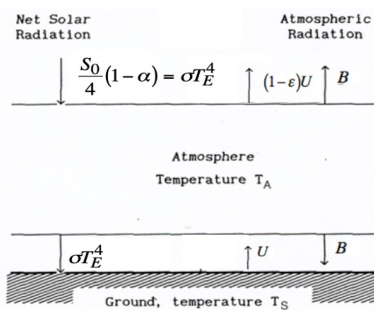
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The Greenhouse Effect

- Case 2:** Assume the atm is semi-grey $\rightarrow \epsilon < 1$



Write the energy balance equations at:

$\rightarrow$ TOA

$\rightarrow$ Atmosphere

$\rightarrow$ Surface

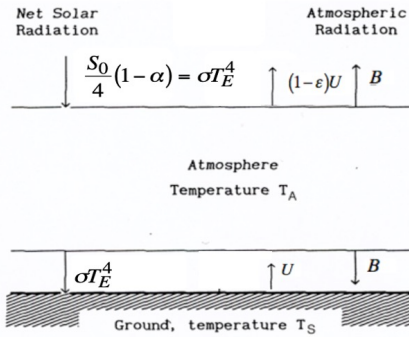
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The Greenhouse Effect

the Kirchhoff's law $\rightarrow$ the atm emits energy at a rate B (upward and downward):



$$B = \epsilon \sigma T_A^4$$

At the surface

$$\frac{S_0}{4}(1-\alpha) + B = U$$

In the atm

$$2B = \epsilon U$$

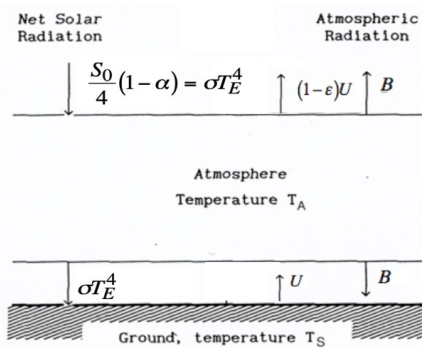
$$\Rightarrow T_S = ?, T_A = ?$$

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The Greenhouse Effect



$$\Rightarrow T_S = \left(\frac{2}{2-\epsilon} \right)^{1/4} T_E$$

$$T_A = \left(\frac{1}{2-\epsilon} \right)^{1/4} T_E$$

$$\epsilon=0.78, \alpha_p=0.3 \rightarrow T_S=288K, T_A=242K$$

$$\epsilon=0.9, \alpha_p=0.3 \rightarrow T_S=296K, T_A=249K$$

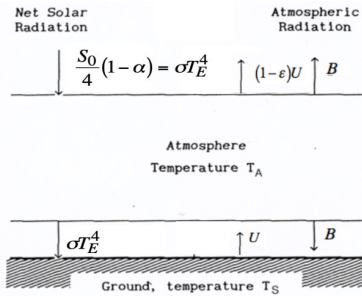
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The Greenhouse Effect

The **strength of greenhouse effect** is measured through the so-called **back radiation**



$$F_{\downarrow} = B = \epsilon \sigma T_A^4 = \frac{\epsilon}{2-\epsilon} \sigma T_E^4 \equiv \frac{\epsilon}{2-\epsilon} Q_{\downarrow}$$

- $\epsilon \uparrow \rightarrow B \uparrow \rightarrow T_s \uparrow$
- The atm may radiate as much energy to the surface as the Sun

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Exercise #1

- **Atm of two layers**
- Supposing the atm is transparent to shortwave radiation; and the atm layers are opaque to longwave radiation ($\epsilon_1 = \epsilon_2 = 1$).
- Surface is considered as a black body.
- The incoming radiation flux is S_0 . The planetary albedo is α
- a) Write the radiation balance equation at each layer;
- b) Calculate surface temperature, and temperature at each atm layer

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Exercise #2

- **Atm of two layers**
- Supposing the atm is transparent to shortwave radiation; and **the emissivity of each layer is ϵ_1 and ϵ_2** , respectively.
- Surface is considered as a black body.
- The incoming radiation flux is S_0 . The planetary albedo is α
- a) **Write the radiation balance equation at each layer;**
- b) **Calculate surface temperature, and temperature at each atm layer**

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? the radiation balance equation
? Variables of the model

At the surface
At Layer #2
At Layer #1
At TOA?

$$T_1 = \sqrt[4]{\frac{2 - \epsilon_2}{A}} T_e$$

$$T_2 = \sqrt[4]{\frac{2 + \epsilon_1 \epsilon_2 - \epsilon_1 \epsilon_2^2}{A}} T_e$$

$$T_g = \sqrt[4]{\frac{4 - \epsilon_1 \epsilon_2}{A}} T_e$$

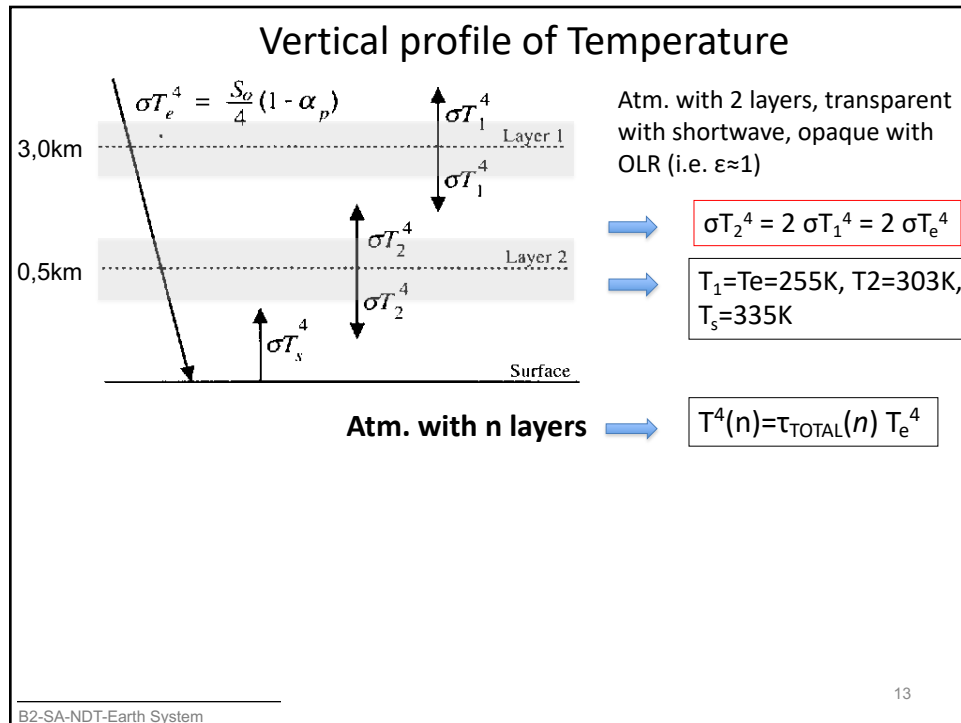
$$A = 4 - 2\epsilon_1 + 2\epsilon_1 \epsilon_2 - 2\epsilon_1 \epsilon_2^2 + \epsilon_1 \epsilon_2^3 - 2\epsilon_2$$

? $T_1 < T_2 < T_g$

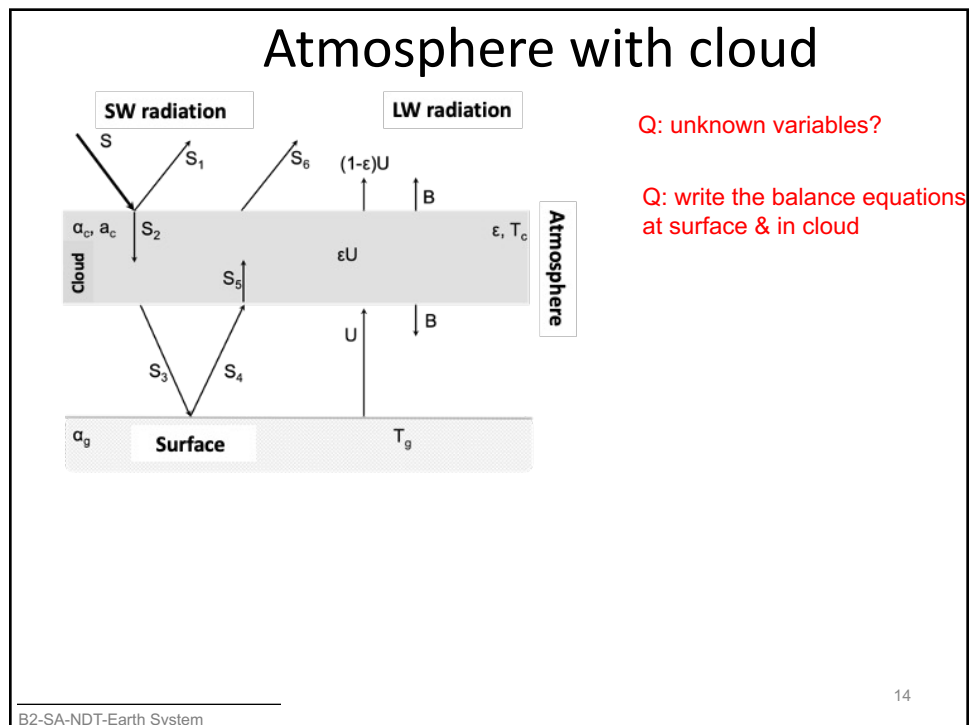
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Atmosphere with cloud

Cloud: α_c, a_c
Atmosphere: ϵ, T_c
Surface: α_g

At surface: $(1 - \alpha_g)S_3 + B = U$

In cloud: $S_2 + S_5 + \epsilon U = 2B$

$$\begin{cases} (1 - \alpha_g)(1 - \alpha_c)(1 - a_c)S + \epsilon \sigma T_c^4 = \sigma T_g^4 \\ a_c(1 - \alpha_c)S + a_c \alpha_g(1 - \alpha_c)(1 - a_c)S + \epsilon \sigma T_g^4 = 2\epsilon \sigma T_c^4 \end{cases}$$

$$\begin{cases} T_g = \sqrt[4]{\frac{(1 - \alpha_c)(2 - a_c)(1 - \alpha_g + \alpha_g a_c)S}{(2 - \epsilon)\sigma}} \\ T_c = \sqrt[4]{\frac{(1 - \alpha_c)S}{\epsilon \sigma} \left[\frac{(2 - a_c)(1 - \alpha_g + \alpha_g a_c)}{(2 - \epsilon)} - (1 - \alpha_g)(1 - a_c) \right]} \end{cases}$$

?

T_g vs. α_c
 T_g vs. a_c
 T_g vs. ϵ

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Practice #4.1 The Greenhouse effect

Net Solar Radiation

$\frac{S_0}{4}(1 - \alpha) = \sigma T_E^4$

Atmospheric Radiation

$(1 - \epsilon)U \uparrow B$

Atmosphere
Temperature T_A

$\sigma T_E^4 \uparrow U \downarrow B$

Ground, temperature T_g

Given:

- Solar constant (S_0) = 1361 W/m²
- Planetary albedo (α_g) = 0.3
- Stefan-Boltzmann constant: 5.67×10^{-8} W/m²K⁴

Write a Python program that calculates and plots the temperatures of the surface & of the atmosphere as a function of the emissivity ϵ , varying from 0 to 1 with the step of 0.01.

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Practice #4.2**Vertical profile of temperature****Given:**

- Solar constant (S_0) = 1361 W/m²
- Planetary albedo (α_g) = 0.3
- Stefan-Boltzmann constant: 5.67×10^{-8} W/m²K⁴

- Write a Python program that simulates an 8-layer atmosphere (n=8)
- Compute the temperature for the surface and each of the 8 atm layers
- Plot the resulting temperatures against their relative heights
- Comment on the results obtained